Sri Sivasubramaniya Nadar College of Engineering

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Title

PAYLOAD-AWARE TWO-TIER UAV ROUTE PLANNING FOR LAST-MILE DELIVERY

REVIEW REPORT – I

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Abstract

UAV systems for medical and emergency last-mile delivery must balance objectives that fundamentally conflict: shorter routes carry higher flight risk, low-risk corridors drain more battery, and energy-efficient paths are rarely the fastest. No single fixed trade-off serves all payload types, yet existing planners apply the same objective weights to every mission regardless of what is being carried.

Existing multi-objective planners—including entropy-based, diversity-driven, and crowding-distance Pareto selection methods—encode objective weights as a fixed scalar before path search begins. Payload properties such as fragility class, temperature sensitivity, and delivery urgency have no influence over which Pareto-optimal route is chosen. When the payload changes, operators must re-tune weights manually and re-run the planner—an intervention that is infeasible under field conditions.

This work proposes a *Payload-Aware Two-Tier UAV Route Planning* system that separates Pareto front construction from route selection, introducing an explicit algebraic bridge between Tier 1 routing objectives and Tier 2 payload metadata. Tier 1 runs a multi-objective A*/Dijkstra graph-search over a delivery airspace graph sourced from OpenStreetMap, producing a Pareto-optimal route set across distance, energy consumption, and flight risk. Tier 2 reads four operator-assigned payload tags and derives objective weights through a fixed formula: fragility class scales w_{risk}, delivery urgency scales w_{dist}, and temperature sensitivity scales w_{energy}. These weights score every Pareto point through a utility function; the arg max is selected as the mission route.

The system will be evaluated on programmatically generated airspace graphs at three scales (50, 200, and 500 nodes) against single-objective A* and MOPSO baselines, using hypervolume, Pareto spread, and Tier 2 selection accuracy as metrics. Hardware validation will be conducted using a Pixhawk-equipped quadcopter through ArduPilot SITL simulation and outdoor flight tests. The principal contribution is a formula-driven mapping from payload metadata to routing objective weights, connecting mission context directly to Pareto point selection without expert intervention.

Keywords: UAV route planning; multi-objective optimisation; Pareto front; payload-aware selection; algebraic bridge; last-mile delivery; ArduPilot

1. Introduction

Unmanned aerial vehicles (UAVs) have emerged as a compelling platform for last-mile delivery in scenarios where ground transportation is impractical—medical supply chains to remote clinics, emergency drug drops during disaster response, and fragile diagnostic sample transport where road vibration risks contamination. In each scenario the UAV must fly a route from a dispatch hub to a delivery point across an airspace that contains obstacles, regulated no-fly zones, and variable wind corridors. A good route balances three conflicting physical objectives simultaneously: minimising total flight distance, minimising battery energy expenditure, and avoiding high-risk segments.

The standard engineering response to such a conflict is to form a weighted sum of the three objectives into a single scalar cost function and then run a conventional path planner such as A* on it. This scalar collapses the multi-objective problem into a single-objective one, and the operator adjusts the three weights to express the mission's priority before each flight. The approach works for a fixed mission type, but it fails immediately when the payload changes: emergency medicine requires a completely different weight profile from fragile biological samples or disaster-relief bulk supplies. Manually re-tuning and re-running the planner between missions is operationally impractical in time-critical field conditions.

1.1 Background

Multi-objective UAV path planning has attracted sustained research attention because no single route can simultaneously minimise distance, energy, and flight risk—these objectives are structurally in conflict. The dominant approach in the literature is to generate a Pareto-optimal front of non-dominated routes, each representing a different trade-off across the objective space, and then select one route from that front as the mission route.

The critical question that determines the practical value of any such system is: *how is the final route selected from the Pareto front?* The three most directly relevant papers reviewed below all answer this question using **front diversity metrics**: entropy, geometric spread, or crowding distance. These metrics select routes that are well-distributed across the Pareto front, but they are entirely blind to the delivery context. None of the three papers allow what the UAV is carrying to influence which route is chosen. This is the structural research gap this project addresses.

Autonomous UAV systems for medical and emergency last-mile delivery oper-

ate under a structural constraint that distinguishes them from general routing problems: the optimal route trade-off is not fixed—it is a function of what is being carried. A fragile diagnostic sample demands a low-vibration, low-turbulence corridor even at the cost of a much longer flight time. Emergency insulin requires the shortest path regardless of risk. Bulk disaster-relief supplies demand the most energy-efficient route to maximise delivery range.

1.2 Motivation

The motivation for this work arises from three real operational scenarios that impose fundamentally incompatible routing requirements:

Fragile medical samples (tissue biopsies, blood samples, pathology specimens) are highly sensitive to vibration, g-forces, and thermal exposure. The dominant routing priority is minimising flight risk—defined here as exposure to turbulent corridors, sudden altitude changes, and wind-exposed segments. A route that adds 40% extra distance but halves the risk of vibration damage is operationally preferable.

Emergency medicine (insulin, epinephrine, antivenom) has a strict time window. The dominant routing priority is minimising route distance to minimise transit time. Flight risk and energy are secondary. For a payload with a 15-minute delivery window, the shortest path wins unconditionally.

Disaster-relief bulk supplies (water purification tablets, bandages, ration packs) weigh significantly more than medical payloads and are not time-critical or fragile. The dominant routing priority is energy efficiency—minimising battery drain to maximise the number of deliveries possible on a single charge cycle.

No existing UAV planner can switch between these three routing strategies based on a payload tag. The proposed two-tier system accomplishes this without any expert intervention or replanning. Payload metadata—fragility class, temperature sensitivity, delivery urgency—has never been used as the sole input to autonomous Pareto point selection in the UAV delivery literature, leaving a structural gap this project directly addresses.

1.3 Objectives

1. To formulate multi-objective UAV route planning as a structured **Pareto front pre-computation** problem over a delivery airspace graph, theoretically distinguishing the deterministic graph-search approach from stochastic population-based alternatives.

2. To design a **payload metadata schema** and a principled rule-based mapping that translates mission context (fragility class, delivery urgency, temperature sensitivity) into Pareto point selection objective weights without any expert weight tuning.

3. To benchmark Tier 1 route quality and Pareto front coverage against single-objective A* and MOPSO baselines on identical delivery scenarios, measuring hypervolume and Pareto spread (Inverted Generational Distance).

4. To quantify **Tier 2 selection accuracy** by measuring alignment between selected route characteristics and payload tag requirements across diverse mission types (fragile medical sample, emergency medicine, disaster-relief supplies).

5. To establish the system's **novelty**: payload metadata is used as the sole input to Pareto point selection, replacing context-blind entropy and diversity-based criteria with a deterministic algebraic mapping from payload description to route choice.

1.4 Scope

This project covers two phases. Phase I delivers a complete, validated software simulation system including the multi-objective A* graph-search engine (Tier 1), the payload-driven route selector (Tier 2), a MOPSO baseline, and a benchmarking suite evaluated on programmatically generated airspace graphs at three scales (50, 200, and 500 nodes). Phase II extends the validated software to physical hardware: a Pixhawk-equipped F450 quadcopter with a Raspberry Pi 5 companion computer, validated through ArduPilot SITL simulation and outdoor flight tests. The project does not address multi-UAV coordination, dynamic obstacle avoidance during flight, or real-time path replanning.

2. Related Work

Multi-objective UAV path planning literature consistently uses Pareto front diversity metrics—entropy, geometric spread, or crowding distance—for route selection, with no regard for the delivery context. Three papers that are most directly relevant to the proposed work are reviewed below. Table 2.1 provides a structured comparison.

Table 2.1: Summary of related work

S.No.	Author/Year	Proposed Method	Key Findings	Limitations
1	Chokkalingam & Ganesan (2026) [1]	Multi-objective UAV path planning with adaptive constraint handling and entropy-based criterion for Pareto front selection.	Strong Pareto front coverage versus MOPSO; entropy criterion selects routes spanning a wider range of objective combinations than random selection.	Payload metadata never incorporated; fragile samples and emergency medicine receive identical route selection from the same front.
2	Datsko et al. (2024) [2]	Diversity-driven Pareto selection for UAV routing that maximises geometric spread of candidate solutions across the multi-objective space.	Improvements in Inverted Generational Distance (IGD) over diversity-agnostic selection; better coverage of the true Pareto front.	Selection is guided entirely by solution diversity; delivery context, payload type, and time window have no influence on which route is chosen.
3	Ntakolia et al. (2023) [3]	Crowding-distance-based Pareto point selection in a multi-objective UAV path planner (NSGA-II style).	Well-spread Pareto fronts for UAV routing; compares favourably against NSGA-II and ε-constraint methods on grid-world airspace environments.	Crowding distance cannot encode urgency, fragility, or thermal sensitivity; mission context is absent from selection at any stage.

2.1 Research Gap

All three reviewed papers select Pareto-optimal routes based on **front diversity metrics**—entropy, geometric spread, and crowding distance respectively. None of them allow the delivery context (what payload is being carried) to influence which route is selected. This is the structural gap the proposed system directly addresses.

The gap is not a minor implementation detail—it is a fundamental design assumption. Entropy, geometric spread, and crowding distance are all defined over the Pareto front itself. They have no interface through which external mission context can enter. Extending these methods to incorporate payload information would require replacing the selection criterion entirely, which is precisely what the proposed Tier 2 utility scoring mechanism does.

Three specific gaps addressed by this work are as follows. **Gap 1 (No payload-aware Pareto selection):** All existing UAV multi-objective planners apply a fixed selection criterion independent of payload type. The proposed Tier 2 introduces a formula-driven mapping from four payload tags directly to Pareto point selection. **Gap 2 (No re-use across mission types):** Existing planners re-run the full multi-objective search when the mission context changes. The proposed two-tier separation pre-computes the Pareto front once (Tier 1) and allows instantaneous route selection for any new payload tag (Tier 2), without re-running the planner. **Gap 3 (No algebraic bridge between delivery context and routing objectives):** The mapping between a payload description and the routing trade-off it implies (e.g., fragility $\rightarrow$ minimise risk, urgency $\rightarrow$ minimise distance) has never been formalised algebraically. The proposed weight-derivation formula provides this bridge, making the mapping deterministic, transparent, and reproducible.

3. Problem Statement

Existing UAV route planners encode distance, energy consumption, and flight risk into a **single hand-tuned weighted-sum scalar** before path search begins, permanently fixing the objective trade-off regardless of what is being carried. When the payload context changes—from fragile diagnostic samples requiring vibration-minimal routing to time-critical emergency medicine requiring shortest-path routing—an expert must manually re-tune the planner weights and re-run the search from scratch. This intervention is **infeasible in field conditions**.

Formally, given: (i) an airspace graph $G = (V, E)$ with three-objective edge cost vectors $\mathbf{c}(e) = [d(e), \varepsilon(e), \rho(e)]$ representing geodesic distance, estimated energy consumption, and flight risk respectively; (ii) a Pareto-optimal front $\mathcal{P}$ of non-dominated routes pre-computed over G; and (iii) a payload metadata tuple $(F, T, U, P_{\text{lvl}})$ encoding fragility class $F \in \{1 \dots 5\}$, temperature sensitivity $T \in \{1 \dots 3\}$, delivery urgency $U \in \{1 \dots 3\}$, and priority level $P_{\text{lvl}} \in \{1 \dots 5\}$—the problem is to select the route $p^* \in \mathcal{P}$ that best satisfies the routing requirements implied by the payload metadata, without expert re-tuning and without re-running the multi-objective search.

The inputs are measurable quantities the operator can assign from domain knowledge (e.g., "this vial is fragile: $F = 5$; delivery is non-urgent: $U = 1$"). The constraints are that selection must be deterministic, reproducible, and complete within one second of payload tag entry. The success criterion is that the selected route characteristics (distance, energy, risk) align with the payload's stated requirements as verified by a Tier 2 selection accuracy metric evaluated across multiple payload presets.

4. Justification for the Problem Formulation

4.1 Need for the Proposed Work

No current UAV route planner adjusts its routing priorities to match the payload being carried. The same fixed objective weights are applied to every mission, forcing field operators to manually recalibrate between every deployment. This is impractical and error-prone in time-critical emergency scenarios. Payload metadata (fragility class, temperature sensitivity, delivery urgency) has never been used as the sole input to autonomous Pareto point selection in the UAV delivery literature, leaving a structural gap this project directly addresses.

The proposed system resolves this by separating Pareto front generation from mission-context-driven route selection. The Pareto front is computed once for a delivery region and stored. For each new mission, only Tier 2 runs—reading the payload tag and selecting the appropriate route from the pre-computed front in near-instantaneous time, with no re-planning and no expert weight adjustment required.

4.2 Feasibility

Technical feasibility: The Tier 1 multi-objective A*/Dijkstra algorithm is a well-studied graph-search extension. The Tier 2 utility scoring is a closed-form algebraic formula executable in microseconds on any hardware. Both components are implementable in Python with standard open-source libraries (NetworkX, NumPy). Hardware deployment on a Raspberry Pi 5 companion computer is feasible given the computational simplicity of Tier 2 route selection ($< 1\,\mathrm{s}$ end-to-end from payload tag input to mission file output).

Operational feasibility: The system requires only four integer tag inputs from the operator, eliminating the need for weight-tuning expertise. The ArduPilot SITL simulation environment allows complete end-to-end validation before outdoor flights, reducing hardware risk to an acceptable level.

Economic feasibility: All software components use open-source libraries (osmnx, NetworkX, NumPy, matplotlib). The hardware platform—F450 quadcopter frame, Pixhawk flight controller, Raspberry Pi 5—uses standard, commercially available components at modest cost. No specialised or proprietary software licences are required.

Schedule feasibility: Phase I (software simulation and benchmarking) is achievable within the first semester aligned to Review I. Phase II (hardware integra-

tion and outdoor testing) is planned for the second semester, with milestones aligned to the Review II and Final Review schedule as shown in Chapter 6.

4.3 Expected Contribution

The principal contribution is a formula-driven mapping from payload metadata to routing objective weights, connecting mission context directly to Pareto point selection without expert intervention. This replaces context-blind diversity criteria with a direct algebraic link from payload description to route choice. The Pareto front is computed once and reused across all mission types without re-running the planner, making the system both operationally efficient and reproducible. The proposed system is the first in the UAV delivery literature to use payload metadata as the *sole* autonomous input to Pareto point selection, constituting a novel algebraic bridge between delivery context and routing objectives.

5. Methodology

The proposed system is built on a single architectural decision: **separate Pareto front construction (Tier 1) from route selection (Tier 2)**. This separation eliminates the need to re-run a multi-objective search each time the payload changes. The Pareto front is computed once for a delivery region and stored persistently. For each new mission, only Tier 2 runs—reading the payload tag and selecting the appropriate route from the pre-computed front in near-instantaneous time.

5.1 Architectural Design for Proposed System

Figure 5.1 shows the complete two-tier system architecture. Airspace graph data from OpenStreetMap enters Tier 1, which performs a multi-objective A*/Dijkstra search and stores the resulting Pareto-optimal route set. For each new mission, Tier 2 reads the four operator-assigned payload tags, derives objective weights algebraically, scores every stored Pareto point via a utility function, and outputs the arg max as the selected mission route along with an ArduPilot-compatible `.waypoints` file.

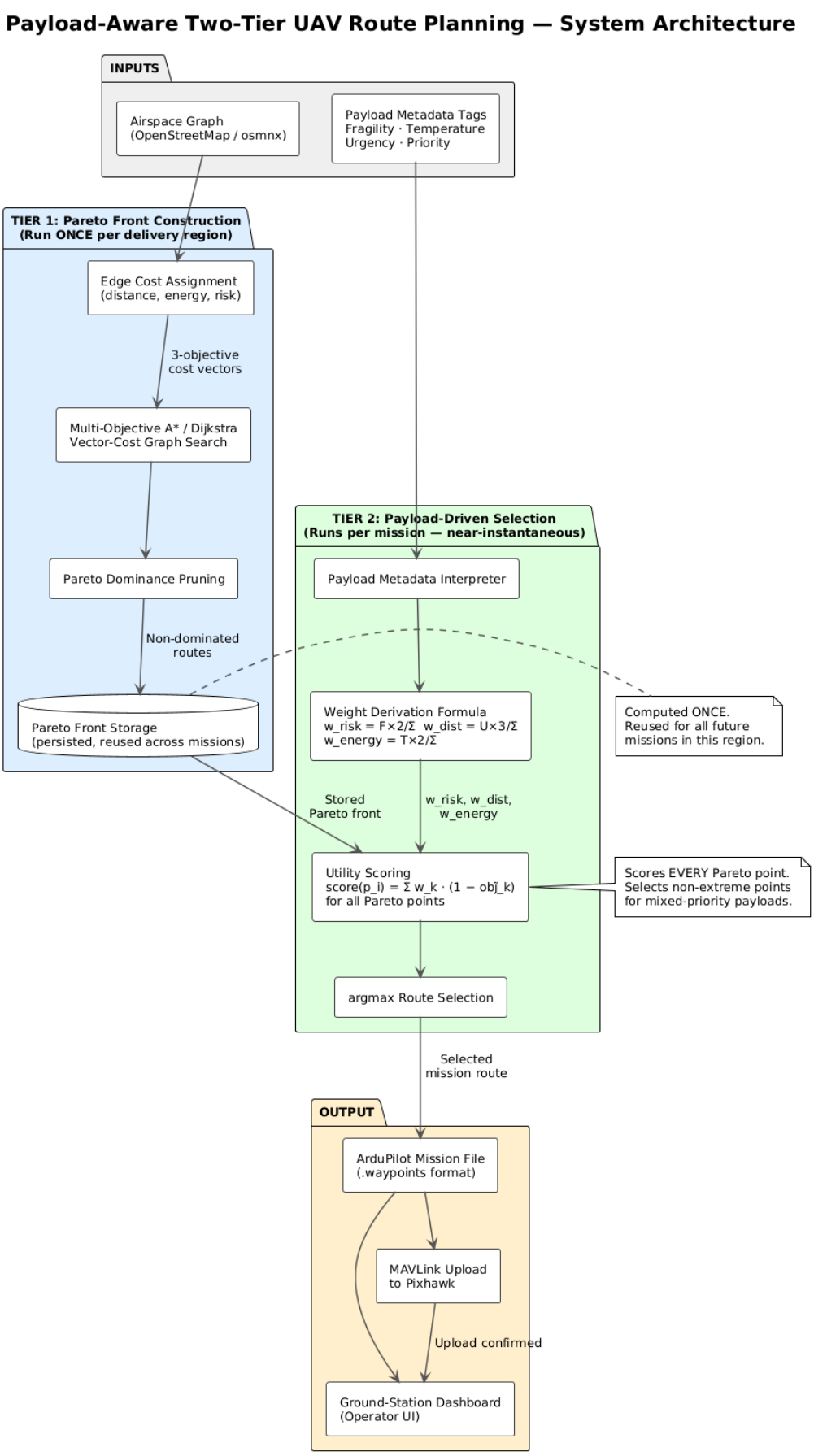


Figure 5.1: Overall system architecture: Two-Tier Payload-Aware UAV Route Planner. Tier 1 pre-computes the Pareto front once per delivery region; Tier 2 selects the optimal route for each new payload tag using the stored front.

The airspace graph $G = (V, E)$ is extracted from OpenStreetMap using the `osmnx` Python library. Each edge $e \in E$ carries a cost vector:

$$\mathbf{c}(e) = \Big[d(e),\ \varepsilon(e),\ \rho(e)\Big]$$

where $d(e)$ is the geodesic distance of the flight segment (km), $\varepsilon(e)$ is the estimated energy consumption (proxy: $d(e)\times$ altitude-change factor), and $\rho(e) \in [0, 1]$ is the flight risk score derived from obstacle density, proximity to no-fly zones, and wind exposure (simulated programmatically for reproducible benchmarking).

The three optimisation objectives are: (1) minimise route distance d; (2) minimise energy consumption ε; and (3) minimise flight risk ρ. A route p dominates route q if p is at least as good as q on all three objectives and strictly better on at least one. The Pareto front $\mathcal{P}$ is the set of all non-dominated routes.

Tier 2 Algebraic Bridge. Each of the four payload metadata tags maps to exactly one Tier 1 routing objective as shown in Table 5.1.

Table 5.1: Semantic mapping: payload metadata tags to Tier 1 routing objectives

Payload Tag	**Range**	**Maps to Tier 1 Objective**	**Reasoning**
Fragility class F	1–5	w_{risk} (flight risk weight)	Fragile items need low-vibration, low-risk corridors
Temperature sensitivity T	1–3	w_{energy} (energy weight)	Temp-sensitive items need short exposure time $\Rightarrow$ fast flight
Delivery urgency U	1–3	w_{dist} (distance weight)	Time-critical deliveries need the shortest path
Priority level P_{lvl}	1–5	Tie-breaker amplifier	Scales the dominant weight when scores are close

From the payload tags, three routing objective weights are derived algorithmically with no expert input:

$$r_{\text{raw}} = F \times 2, \quad d_{\text{raw}} = U \times 3, \quad e_{\text{raw}} = T \times 2 \tag{5.1}$$

$$w_{\text{risk}} = \frac{r_{\text{raw}}}{r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}}, \quad w_{\text{dist}} = \frac{d_{\text{raw}}}{r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}}, \quad w_{\text{energy}} = \frac{e_{\text{raw}}}{r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}} \tag{5.2}$$

The multipliers (2 for fragility and temperature, 3 for urgency) reflect the domain insight that urgency has a higher penalty sensitivity than fragility or thermal constraints

in the UAV medical delivery context.

For each Pareto point p_i with objective values $(d_i, \varepsilon_i, \rho_i)$, min-max normalised values $\tilde{d}_i$, $\tilde{\varepsilon}_i$, $\tilde{\rho}_i$ are computed over the full Pareto front $\mathcal{P}$. The utility score is:

$$\text{score}(p_i) = w_{\text{dist}} \cdot (1 - \tilde{d}_i) + w_{\text{energy}} \cdot (1 - \tilde{\varepsilon}_i) + w_{\text{risk}} \cdot (1 - \tilde{\rho}_i) \tag{5.3}$$

The selected mission route is $p^* = \arg\max_{p_i \in \mathcal{P}} \text{score}(p_i)$. For high-priority missions ($P_{\text{lvl}} \geq 4$), the dominant weight is amplified by $0.1 \times w_{\max}$ to sharpen the selection.

The Tier 1 and Tier 2 activity flows are shown in Figures 5.2 and 5.3 respectively.

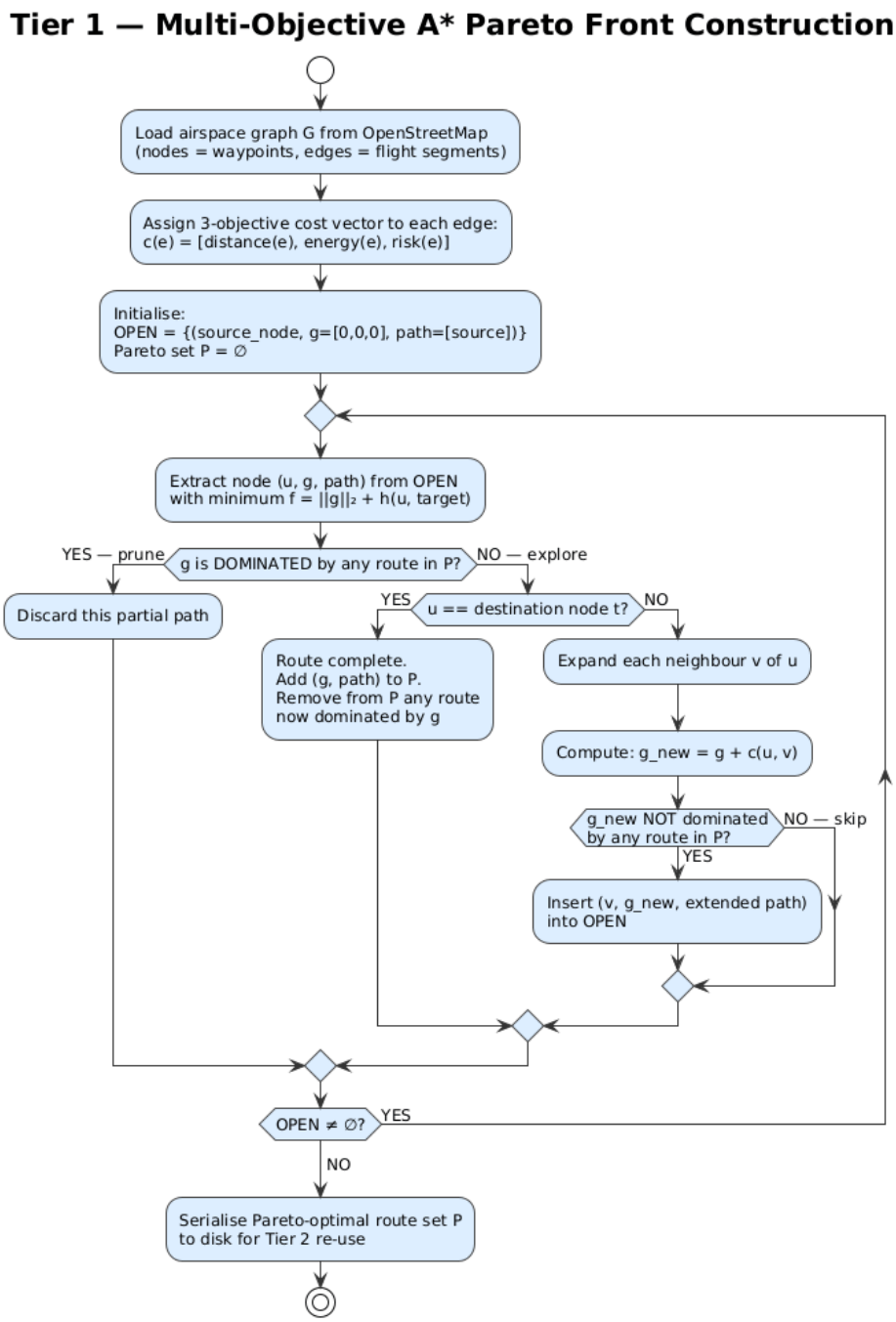


Figure 5.2: Tier 1 activity flow: Multi-Objective A* Pareto Front Construction. The open set is explored in order of estimated total cost until no further non-dominated routes exist.

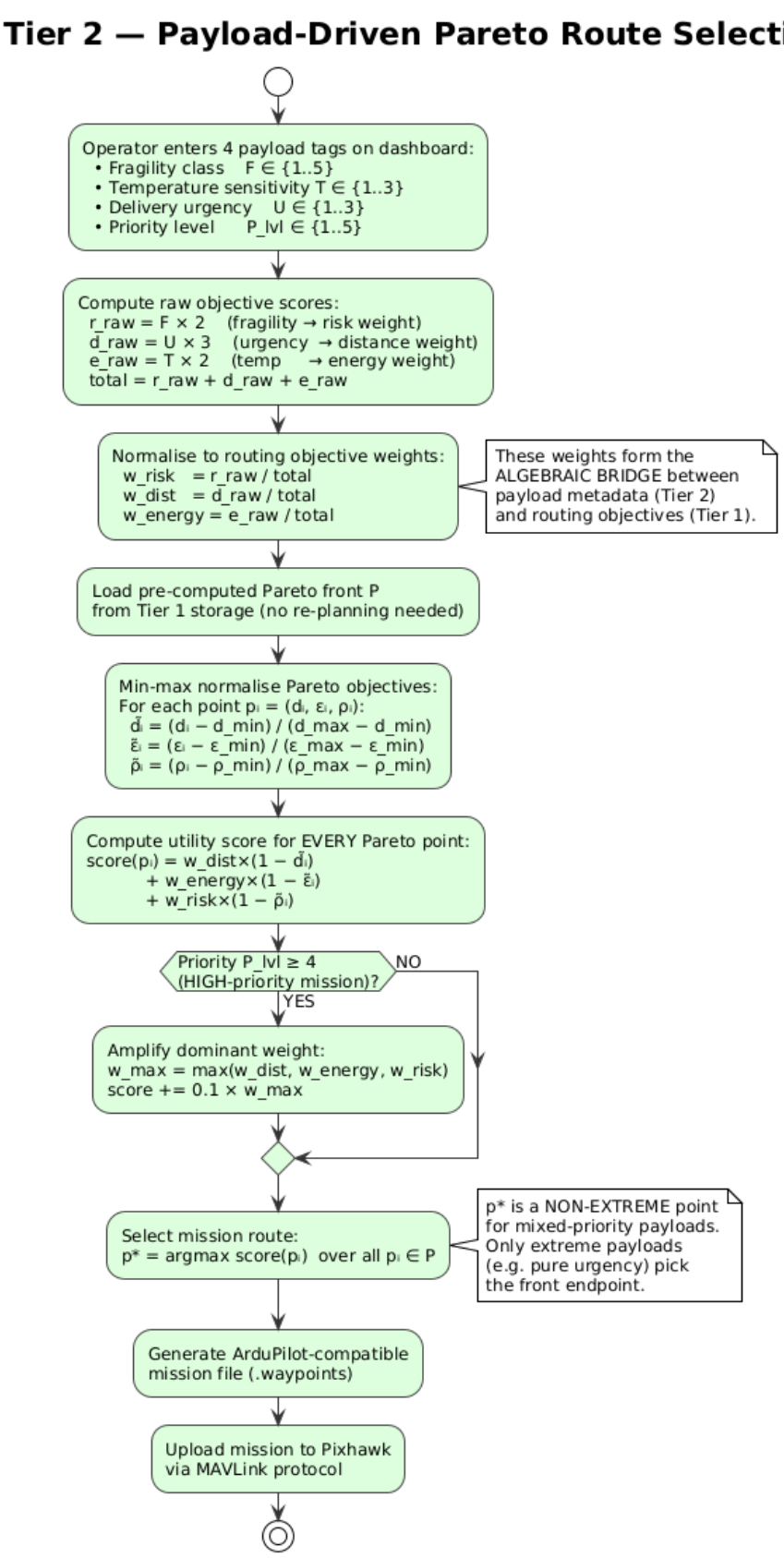


Figure 5.3: Tier 2 activity flow: Payload-Driven Pareto Selection. Payload tags are algebraically mapped to weights, which score every Pareto point via a utility function; the arg max is selected as the mission route.

5.2 Modules

5.2.1 Module 1: Airspace Graph Builder

Extracts the road/path network from OpenStreetMap for a specified delivery region using `osmnx`. Assigns three objective costs to each edge: geodesic distance, energy (distance $\times$ altitude-change factor), and flight risk (simulated threat zones and obstacle density scored 0–1 using a configurable hazard map). Output: a directed weighted graph G stored as a serialised NetworkX object ready for Tier 1 search.

5.2.2 Module 2: Multi-Objective A*/Dijkstra Engine (Tier 1)

Implements the multi-objective A* algorithm maintaining a cost vector $\mathbf{g}$ (not a scalar) for each partial path. Paths dominated by already-discovered routes are pruned early. The `UpdateParetoFront` subroutine adds a new route and removes any previously stored routes it dominates. Output: a deterministic, reproducible Pareto front $\mathcal{P}$ serialised to disk for Tier 2 re-use across all future missions in the same delivery region.

5.2.3 Module 3: Payload Metadata Interpreter (Tier 2)

Implements the payload-driven route selection algorithm: reads the four payload tags, derives objective weights via Equations (5.1)–(5.2), applies min-max normalisation to Pareto objectives, computes a utility score for every Pareto point via Equation (5.3), and returns the arg max route. Optionally amplifies the dominant weight for high-priority missions ($P_{\text{lvl}} \geq 4$). Generates an ArduPilot-compatible `.waypoints` mission file for upload.

5.2.4 Module 4: Testing and Evaluation

Evaluates all methods on three scenario scales: S1 ($\approx$50 nodes, low obstacle density), S2 ($\approx$200 nodes, medium density), and S3 ($\approx$500 nodes, high density). The MOPSO baseline runs with swarm size 100, 200 iterations, and a Pareto archive of 100 solutions, repeated 30 times per scenario to obtain mean $\pm$ standard deviation. Metrics include Hypervolume (HV), Inverted Generational Distance (IGD), and Tier 2 selection accuracy across four payload presets. Phase II testing extends to ArduPilot SITL validation and outdoor flight tests on an F450 quadcopter with Pixhawk flight controller, measuring route adherence (GPS position error vs. planned route) and end-to-end latency.

6. Project Timeline Chart

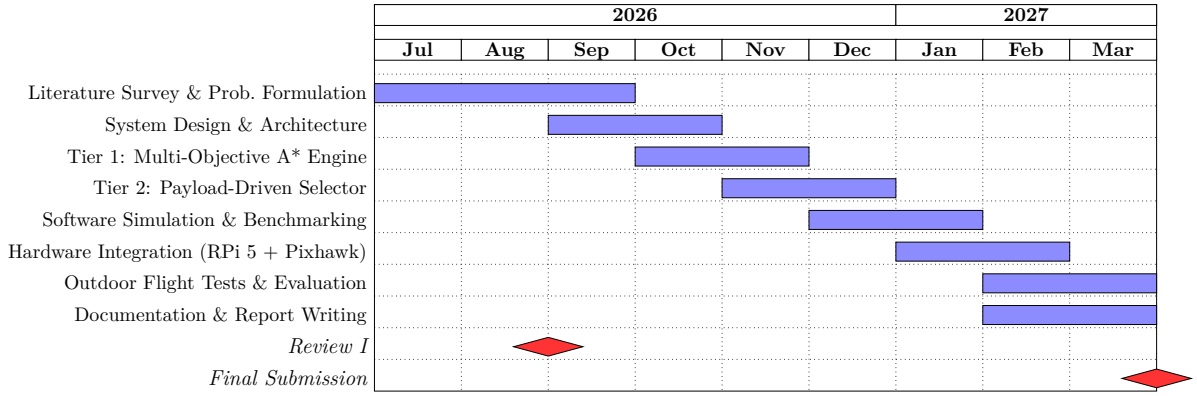


Figure 6.1: Project timeline (July 2026 – March 2027). Red diamonds mark review and submission milestones.

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